

# Technical Assignment (With Plivo API)

## Position Overview

We are seeking a Forward Deployed Engineer (FDE) to join our team to build communication-forward applications using APIs like Plivo. This assignment evaluates your ability to integrate APIs, handle call logic, and design a multi-level IVR.

## Technical Assignment

### Scenario

InspireWorks wants a demo IVR system that demonstrates Plivo's voice API capabilities. The system should:

1. Make an outbound call to a given phone number.
2. Upon answer, present a multi-level IVR menu:
  - a. Level 1: Language selection
    - i. Press 1 for English
    - ii. Press 2 for Spanish
  - b. Level 2 (after language selection):
    - i. Press 1 to play a short audio message
    - ii. Press 2 to connect to a live associate (use a placeholder number)

The goal is to demonstrate basic voice interaction, branching logic, and Plivo API integration.

## Implementation Requirements

1. **Outbound Call**
  - Build an endpoint or script to initiate an outbound call using Plivo's API.
  - Target phone number can be provided via a simple UI or configuration variable.
2. **IVR Menu**
  - Level 1: Prompt caller to select a language (English / Spanish).
  - Level 2: Depending on language choice:
    - Press 1 → Play an audio file (e.g., MP3 hosted publicly).
    - Press 2 → Forward call to an associate (can be a placeholder number).

- Use Plivo XML (or equivalent JSON API) for call flow.
3. **Multi-level Flow Handling**
    - Correctly handle DTMF input for multiple levels.
    - Implement branching logic based on user input.
    - Handle invalid inputs gracefully (e.g., repeat the menu).
  4. **Optional Frontend**
    - Simple web page or CLI to trigger the outbound call.
    - Not required, but helpful for demonstration.

## Deliverables

1. **Working Application**
  - Script / web app that triggers outbound calls and supports IVR.
  - Demonstrate Level 1 and Level 2 interactions.
2. **Code Repository**
  - GitHub or zip file with source code.
  - Include README with:
    - Setup instructions
    - Required Plivo credentials (Auth ID / Auth Token)
    - Steps to run and test
3. **Short Demo Video** (3–5 min)
  - Show a call being made
  - Navigate Level 1 → Level 2 menus
  - Demonstrate audio playback and call forwarding

## Evaluation Criteria

## Notes / Tips

- For audio playback, use a publicly accessible MP3 file or Plivo's default audio.
- The live associate number can be a placeholder or your own test number.